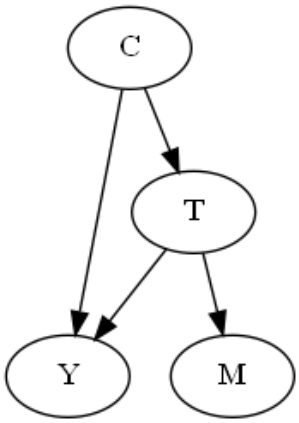
1. Exact and Approximate Inference: ..... *Total: 22 marks*

(1) Dr. Johnson is a researcher who studies a special mental health disorder and its possible treatments. He has two different treatments, called A and B, that he has been testing on his patients for a long time. He has collected thousands of data records from his patients, each recording the treatment a patient received, $T \in \{A, B\}$, and whether the treatment was effective for that patient, $Y \in \{yes, no\}$. He has also observed that his patients vary in how long they stay in negative thinking patterns. He introduces a variable C and labels the duration of negative thinking as either *short* or *long* in the records. The variable C can influence both which treatment a patient receives and the effect of that treatment. He has also noticed that his patients have different levels of social isolation. For that, he introduces a variable M, influenced by the treatment, and labels the data records as either *severe* or *mild*, depending on the extent of the isolation.

(5 marks) (a). Create a DAG for a Bayesian network based on the above description.

Solution:



Edges: $C \to T$, $C \to Y$, $T \to Y$, $T \to M$.

(2 marks) (b). Write the joint distribution of all variables and factorise it over the DAG.

Solution: $P(C, T, M, Y) = P(C)\, P(T \mid C)\, P(M \mid T)\, P(Y \mid C, T)$.

(4 marks) (c). List the steps to sample a full particle from the Bayesian network using forward sampling. Is the order of sampling unique and why? Hint: Use uppercase letters to denote random variables, and lowercase letters to denote values of these variables.

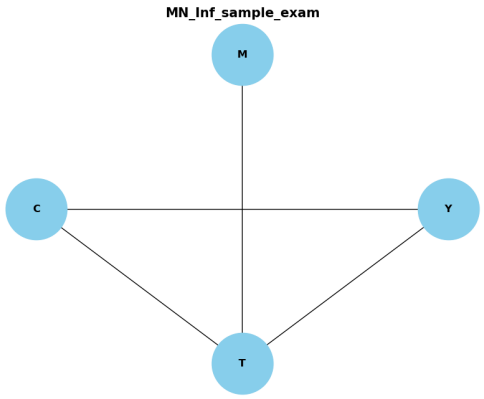
Solution:

1. Sample c from $P(C)$
2. Sample t from $P(T \mid c)$
3. Sample m from $P(M \mid t)$
4. Sample y from $P(Y \mid c, t)$

Return (c, t, m, y).

The order is not unique. Forward sampling only requires a topological order of the DAG, so that every variable is sampled after its parents. C must come first and T second, but M and Y are not ancestors of each other, so Steps 3 and 4 can be swapped.

(4 marks) (d). Transform the above Bayesian network to a Markov network and explain the key ideas behind this.



Solution:

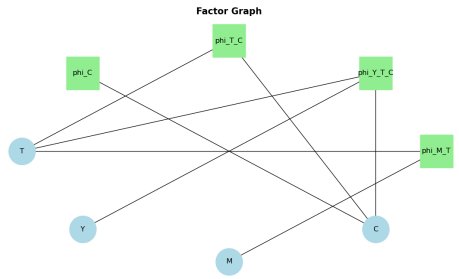
Edges: $C - T$, $C - Y$, $T - Y$, $T - M$.

Moralisation transforms a Bayesian network into a Markov network in two steps: (1) for every node, connect all pairs of its parents with an undirected edge ("marry" the parents); (2) drop the direction of all edges. This guarantees that the scope of every CPD $P(X \mid \mathrm{Pa}_X)$ forms a clique, so each CPD can serve as a clique factor in the Markov network.

In this DAG the only node with more than one parent is Y, and its parents C and T are already connected by $C \to T$. There is therefore no immorality, no edge needs to be added, and it suffices to convert all directed edges to undirected edges.

(4 marks) (e). Transform the above Markov network to a factor graph, using one factor per CPD of the Bayesian network, and write down the Gibbs distribution for

$P(C, T, M, Y)$.



Solution:

Factor nodes and their neighbours: $\phi(C)$–$\{C\}$; $\phi(C,T)$–$\{C,T\}$; $\phi(C,T,Y)$–$\{C,T,Y\}$; $\phi(T,M)$–$\{T,M\}$.

$P(C,T,M,Y) = \frac{1}{Z}\,\phi(C)\cdot\phi(C,T)\cdot\phi(T,M)\cdot\phi(C,T,Y)$,

where $\phi(C) = P(C)$, $\phi(C,T) = P(T \mid C)$, $\phi(T,M) = P(M \mid T)$ and $\phi(C,T,Y) = P(Y \mid C,T)$. Because every factor is a CPD, $Z = 1$.

(3 marks) (f). Derive $P(Y \mid M = mild)$ symbolically by completing the remaining steps of the Sum-Product (variable elimination) algorithm, ensuring each step is specific to this Bayesian network, as shown in the Week 9 seminar slides for the Student example. Hint: use $\phi(X, Z)$ to denote an initial factor and use $\phi(X)[Z = z]$ to denote a factor over variable X reduced by the evidence $Z = z$.

The elimination order is C, T.
1. Eliminating C:
$\psi_1(C,T,Y) = \phi(C)\cdot\phi(C,T)\cdot\phi(C,T,Y)$
$\tau_1(T,Y) = \sum_C \psi_1(C,T,Y)$

Solution: 2. Reducing the factor that involves the evidence and eliminating T:
$\psi_2(T,Y)[M = mild] = \tau_1(T,Y)\cdot\phi(T)[M = mild]$
$\tau_2(Y)[M = mild] = \sum_T \psi_2(T,Y)[M = mild]$

3. Normalising, since $\tau_2(Y)[M = mild] = P(Y, M = mild)$ is unnormalised:
$$P(Y \mid M = mild) = \frac{\tau_2(Y)[M = mild]}{\sum_y \tau_2(y)[M = mild]}$$